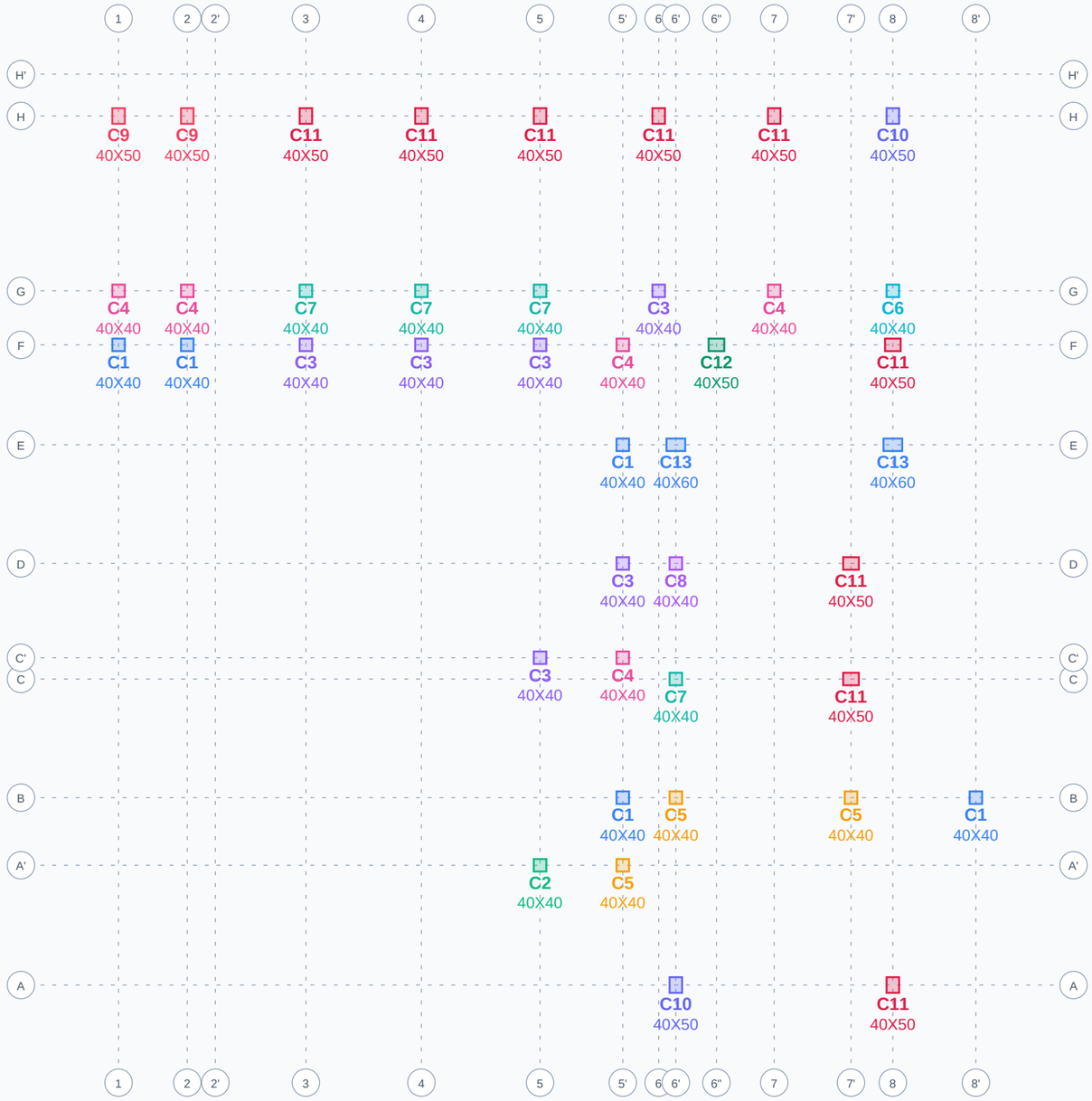
ETABS-to-CAD Column Rebar Reconciliation & Detailer Suggester

Structural Auditing and Optimization Schedule Report

Target ETABS Model:	Rest House Madayi - 2026.01.13.EDB
AutoCAD Drawing Database:	Column reinforcement.dwg (Deduplicated Schedule)
Rebar Safety Buffer:	0.00%
Total Concrete Columns Audited:	42
Conforming Matches suggested:	42 / 42 columns (100.0%)
Non-conforming (Exceeded) warnings:	0
Section Size Mismatch warnings:	0
Average Suggestion Efficiency Index:	**90.84%** (Economical Optimization)
Date of Report Generation:	2026-06-02 12:03:10

Interactive Structural Plan Layout & Groups Summary



Group	CAD Name	Size	Rebar Details	Qty
C1	C-4x4-1	400x400	Main-8 #16	5
C2	C-4x4-2	400x400	Main- 4 #20 +4 #16	1
C3	C-4x4-4	400x400	Main-8 #20	6
C4	C-4x4-6	400x400	Main- 4 #20 +8 #16	5
C5	C-4x4-7	400x400	Main - 4 #25+4 #20	3
C6	C-4x4-8	400x400	Main - 4 #25 +8 #16	1
C7	C-4x4-10	400x400	Main - 8 #25	4
C8	C-4x4-11	400x400	Main- 4 #25 + 8#20	1
C9	C-4x5-2	400x500	Main- 4 #20+4 #16	2
C10	C-4x5-4	400x500	Main- 8 #20	2
C11	C-4x5-8	400x500	Main- 4 #25+8 #16	9
C12	C-4x5-16	400x500	Main-12 #25	1
C13	C-4x6-8	400x600	Main- 4 #25+8 #20	2

Structural Optimization & Audit Executive Summary

This report reconciles the required reinforcement percentages computed by CSI ETABS concrete design with the standard column detailed options available inside the AutoCAD drawing schedule. The suggested columns represent the most cost-effective and steel-economical conforming choices, checking dimensions in both orientation angles. Columns marked with **REINFORCEMENT EXCEEDED** require either cross-section size expansions or custom detailed rebar schedules, as their required reinforcement exceeds the drawing limits.

Total Columns	Conforming	Exceeded	Average Efficiency
42	**42**	0	**90.8%**

Column Mapping & Detail Selection Schedule

Notation	Story Range	Column Label	Grid	ETABS Size	Req %	CAD Name	CAD %	Rebar Details	Eff %
C1	FF to Terrace	C5	F-1	400x400	1.00%	**C-4x4-1**	1.00%	Main-8 #16	**100.0%**
C1	FF to Terrace	C6	F-2	400x400	0.89%	**C-4x4-1**	1.00%	Main-8 #16	**89.0%**
C1	FF to Terrace	C24	E-5'	400x400	0.88%	**C-4x4-1**	1.00%	Main-8 #16	**88.0%**
C1	FF to SF	C32	B-5'	400x400	0.92%	**C-4x4-1**	1.00%	Main-8 #16	**92.0%**
C1	FF to SF	C43	B-8'	400x400	0.85%	**C-4x4-1**	1.00%	Main-8 #16	**85.0%**
C2	FF to GF	C39	A'-5	400x400	1.25%	**C-4x4-2**	1.29%	Main- 4 #20 +4 #16	**96.9%**
C3	FF to Terrace	C17	G-6	400x400	1.56%	**C-4x4-4**	1.57%	Main-8 #20	**99.4%**
C3	FF to Terrace	C9	F-3	400x400	1.38%	**C-4x4-4**	1.57%	Main-8 #20	**87.9%**
C3	FF to Terrace	C12	F-4	400x400	1.37%	**C-4x4-4**	1.57%	Main-8 #20	**87.3%**
C3	FF to Terrace	C15	F-5	400x400	1.39%	**C-4x4-4**	1.57%	Main-8 #20	**88.5%**
C3	FF to SF	C37	D-5'	400x400	1.50%	**C-4x4-4**	1.57%	Main-8 #20	**95.5%**
C3	FF to GF	C35	C'-5	400x400	1.37%	**C-4x4-4**	1.57%	Main-8 #20	**87.3%**
C4	FF to Terrace	C3	G-1	400x400	1.61%	**C-4x4-6**	1.79%	Main- 4 #20 +8 #16	**89.9%**
C4	FF to Terrace	C4	G-2	400x400	1.79%	**C-4x4-6**	1.79%	Main- 4 #20 +8 #16	**100.0%**
C4	FF to Terrace	C20	G-7	400x400	1.66%	**C-4x4-6**	1.79%	Main- 4 #20 +8 #16	**92.7%**
C4	FF to Terrace	C16	F-5'	400x400	1.60%	**C-4x4-6**	1.79%	Main- 4 #20 +8 #16	**89.4%**
C4	FF to SF	C34	C'-5'	400x400	1.58%	**C-4x4-6**	1.79%	Main- 4 #20 +8 #16	**88.3%**
C5	FF to SF	C31	B-6'	400x400	1.88%	**C-4x4-7**	2.00%	Main - 4 #25+4 #20	**94.0%**
C5	FF to SF	C30	B-7'	400x400	1.99%	**C-4x4-7**	2.00%	Main - 4 #25+4 #20	**99.5%**

Notation	Story Range	Column Label	Grid	ETABS Size	Req %	CAD Name	CAD %	Rebar Details	Eff %
C5	FF to GF	C40	A'-5'	400x400	1.85%	**C-4x4-7**	2.00%	Main - 4 #25+4 #20	**92.5%**
C6	FF to Terrace	C22	G-8	400x400	2.14%	**C-4x4-8**	2.23%	Main - 4 #25 +8 #16	**96.0%**
C7	FF to Terrace	C8	G-3	400x400	2.42%	**C-4x4-10**	2.45%	Main - 8 #25	**98.8%**
C7	FF to Terrace	C11	G-4	400x400	2.38%	**C-4x4-10**	2.45%	Main - 8 #25	**97.1%**
C7	FF to Terrace	C14	G-5	400x400	2.40%	**C-4x4-10**	2.45%	Main - 8 #25	**98.0%**
C7	FF to SF	C33	C-6'	400x400	2.44%	**C-4x4-10**	2.45%	Main - 8 #25	**99.6%**
C8	FF to SF	C38	D-6'	400x400	2.49%	**C-4x4-11**	2.80%	Main- 4 #25 + 8#20	**89.1%**
C9	FF to Terrace	C1	H-1	400x500	0.85%	**C-4x5-2**	1.03%	Main- 4 #20+4 #16	**82.5%**
C9	FF to Terrace	C2	H-2	400x500	0.90%	**C-4x5-2**	1.03%	Main- 4 #20+4 #16	**87.4%**
C10	FF to Terrace	C21	H-8	400x500	1.17%	**C-4x5-4**	1.26%	Main- 8 #20	**92.9%**
C10	FF to GF	C41	A-6'	400x500	1.23%	**C-4x5-4**	1.26%	Main- 8 #20	**97.6%**
C11	FF to Terrace	C7	H-3	400x500	1.40%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**78.2%**
C11	FF to Terrace	C10	H-4	400x500	1.30%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**72.6%**
C11	FF to Terrace	C13	H-5	400x500	1.61%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**89.9%**
C11	FF to Terrace	C18	H-6	400x500	1.39%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**77.7%**
C11	FF to Terrace	C19	H-7	400x500	1.55%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**86.6%**
C11	FF to Terrace	C23	F-8	400x500	1.68%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**93.9%**
C11	FF to SF	C28	D-7'	400x500	1.55%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**86.6%**
C11	FF to SF	C29	C-7'	400x500	1.56%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**87.2%**
C11	FF to GF	C42	A-8	400x500	1.40%	**C-4x5-8**	1.79%	Main- 4 #25+8 #16	**78.2%**
C12	FF to Terrace	C44	F-6''	400x500	2.78%	**C-4x5-16**	2.94%	Main-12 #25	**94.4%**
C13	FF to Terrace	C25	E-6'	400x600	1.81%	**C-4x6-8**	1.86%	Main- 4 #25+8 #20	**97.3%**
C13	FF to Terrace	C26	E-8	400x600	1.69%	**C-4x6-8**	1.86%	Main- 4 #25+8 #20	**90.9%**